

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

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Chemistry

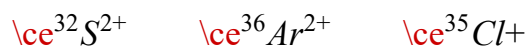
- 1 Equal volumes of $0.100 \text{ mol dm}^{-3}$ HCl(aq) (a strong acid) and $0.100 \text{ mol dm}^{-3}$ $\text{CH}_3\text{COOH(aq)}$ (a weak acid) are investigated in separate experiments at 25°C .

Which of the following statements is/are correct?

- 1 The initial rate of reaction when excess solid calcium carbonate is added is greater with HCl(aq) than with $\text{CH}_3\text{COOH(aq)}$.
- 2 When excess magnesium is added to each acid solution, the total volume of hydrogen gas produced (measured at room temperature and pressure) after both reactions have finished is greater with HCl(aq) than with $\text{CH}_3\text{COOH(aq)}$.
- 3 The volume of $0.0500 \text{ mol dm}^{-3}$ $\text{Ba(OH)}_2\text{(aq)}$ required to completely neutralise the acid solution is the same for both acids.

- A none of them
 B 1 only
 C 2 only
 D 3 only
 E 1 and 2 only
 F 1 and 3 only
 G 2 and 3 only
 H 1, 2 and 3

- 2 Three positive ions are detected in a mass spectrometer:



Which of the following statements is/are correct?

- 1 $^{36}\text{Ar}^{2+}$ and $^{35}\text{Cl}^{+}$ have the same electron configuration.
- 2 $^{36}\text{Ar}^{2+}$ has more neutrons than $^{35}\text{Cl}^{+}$.
- 3 $^{32}\text{S}^{2+}$ has the lowest mass-to-charge ratio (m/z) of the three ions.

- A none of them
 B 1 only
 C 2 only
 D 3 only
 E 1 and 2 only
 F 1 and 3 only
 G 2 and 3 only
 H 1, 2 and 3

3 Four metals W, X, Y and Z have the following properties.

- W displaces X from aqueous X sulfate.
- X reacts with dilute hydrochloric acid, but Y does not.
- Z is extracted from its molten compound by electrolysis.
- W can be extracted from its oxide by reduction with carbon.

Which order shows decreasing reactivity?

- A** $Z > W > X > Y$
B $Z > X > W > Y$
C $W > Z > X > Y$
D $Z > W > Y > X$
E $W > X > Z > Y$

4 In each of the following four procedures, an excess of an aqueous reagent is added to 1.0 dm^3 of a 1.0 mol dm^{-3} aqueous solution of a salt:

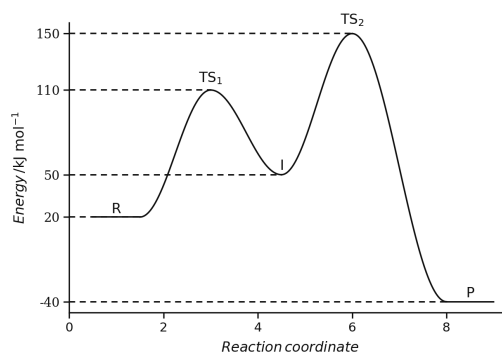
- excess aqueous silver nitrate added to magnesium chloride solution
- excess aqueous barium chloride added to aluminium sulfate solution
- excess aqueous sodium hydroxide added to copper(II) sulfate solution
- excess aqueous sodium chloride added to potassium nitrate solution

Which row in the following table identifies the procedures that form a precipitate and produce the largest, and the smallest, theoretical mass of precipitate?

(M_r values: $\text{AgCl} = 143.5$; $\text{BaSO}_4 = 233$; $\text{Cu(OH)}_2 = 98$; $\text{KCl} = 74.5$; $\text{NaNO}_3 = 85$)

	Procedure that produces the largest mass of precipitate	Procedure that produces the smallest mass of precipitate
A	AgNO_3 and MgCl_2	NaOH and CuSO_4
B	AgNO_3 and MgCl_2	NaCl and KNO_3
C	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$	AgNO_3 and MgCl_2
D	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$	NaOH and CuSO_4
E	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$	NaCl and KNO_3
F	NaOH and CuSO_4	BaCl_2 and $\text{Al}_2(\text{SO}_4)_3$

- 5 The reaction profile for a two-step uncatalysed reaction $R \rightarrow I \rightarrow P$ is shown below. In the presence of a catalyst, the activation energy of the forward rate-determining step is reduced by 30 kJ mol^{-1} , while the activation energy of the other step is unaffected. What is the activation energy of the rate-determining step for the reverse catalysed reaction $P \rightarrow R$?



- A 60 kJ mol^{-1}
 B 70 kJ mol^{-1}
 C 100 kJ mol^{-1}
 D 130 kJ mol^{-1}
 E 160 kJ mol^{-1}
 F 190 kJ mol^{-1}
- 6 Which of the following reactions involve(s) oxidation?
- 1 $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
 2 $\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$
 3 $2\text{CrO}_4^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 4 $\text{SO}_2(\text{g}) + 2\text{H}_2\text{S}(\text{g}) \rightarrow 3\text{S}(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- A 1 and 2 only
 B 1 and 3 only
 C 1 and 4 only
 D 2 and 3 only
 E 2 and 4 only
 F 3 and 4 only
 G 1, 2 and 4 only
 H 1, 3 and 4 only

- 7 The table shows some physical properties of four pure solid substances, W, X, Y, and Z.

Substance	Melting point / °C	Electrical conductivity as a solid	Electrical conductivity when molten	Electrical conductivity of aqueous solution
W	772	poor	good	good
X	1650	poor	poor	insoluble
Y	122	poor	poor	good
Z	146	poor	poor	poor

Consider the following statements:

1. The electrical conductivity of molten W is due to the movement of delocalised electrons.
2. Melting substance X involves breaking strong covalent bonds.
3. When substance Y is added to water, mobile ions are produced.

Which of the statements is/are correct?

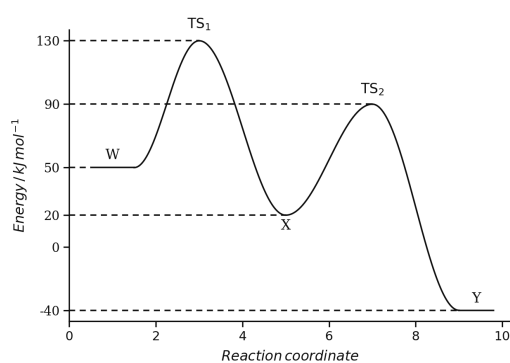
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

8 The reaction energy profile for the two-step reaction $W \rightarrow X \rightarrow Y$ is shown below.

Which row in the table gives the correct values for:

- the overall enthalpy change (ΔH) for the forward reaction $W \rightarrow Y$;
- the activation energy (E_a) of the rate-determining step for the forward reaction;
- the activation energy (E_a) of the rate-determining step for the reverse reaction $Y \rightarrow W$?

	ΔH (overall forward) / $\text{kJ}\cdot\text{mol}^{-1}$	E_a (forward RDS) / $\text{kJ}\cdot\text{mol}^{-1}$	E_a (reverse RDS) / $\text{kJ}\cdot\text{mol}^{-1}$
A	-90	+80	+130
B	-90	+80	+170
C	-90	+70	+110
D	-90	+130	+130
E	+90	+80	+130
F	+90	+80	+170
G	+90	+70	+110
H	+90	+130	+170



9 In this question, all aqueous solutions have the same concentration in mol dm^{-3} .

Three separate experiments were carried out at room temperature by mixing equal volumes of the following solutions:

experiment 1: $\text{MgSO}_4(\text{aq})$ and $\text{Ba}(\text{OH})_2(\text{aq})$

experiment 2: $\text{Na}_2\text{SO}_4(\text{aq})$ and $\text{Ba}(\text{OH})_2(\text{aq})$

experiment 3: $\text{K}_2\text{SO}_4(\text{aq})$ and $\text{Ba}(\text{OH})_2(\text{aq})$

For each experiment, the resulting mixture was filtered, and the solid residue was washed, dried, and weighed.

Which of the following statements is/are correct?

1 Experiment 1 produces a greater mass of dry solid than Experiment 2.

2 Experiment 3 produces a greater mass of dry solid than Experiment 2.

3 After removing the solid, the filtrate from Experiment 1 has a lower concentration of dissolved ions than the filtrate from Experiment 2.

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

10 Three separate aqueous solutions are electrolysed using inert electrodes:

- Solution 1: concentrated NaCl(aq)
- Solution 2: dilute $\text{H}_2\text{SO}_4\text{(aq)}$
- Solution 3: concentrated $\text{CuSO}_4\text{(aq)}$

The same quantity of electrical charge is passed through each solution. Assume that all gases evolved behave ideally and are completely insoluble in water.

Which of the following statements is/are correct?

- 1 The volume of gas evolved at the cathode in Solution 1 is equal to the volume of gas evolved at the cathode in Solution 2 (measured at the same temperature and pressure).
- 2 The volume of gas evolved at the anode in Solution 2 is twice the volume of gas evolved at the cathode in Solution 2.
- 3 As electrolysis proceeds, the pH of Solution 1 increases and the pH of Solution 3 decreases.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

11 The equation summarises the oxidation of iodine by concentrated nitric acid:



What values of w and y are needed to balance the equation?

	w	y
A	6	2
B	8	3
C	10	4
D	10	5
E	12	4

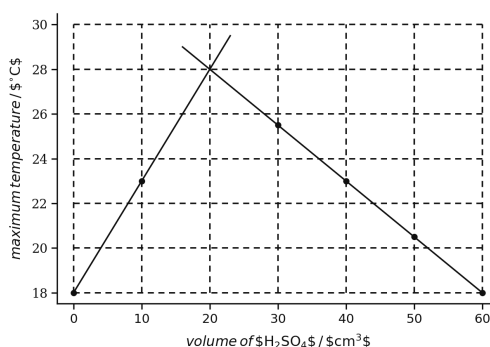
- 12 In a series of experiments to determine the molar enthalpy change of neutralisation, various volumes of $1.0 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4(\text{aq})$ and $1.0 \text{ mol dm}^{-3} \text{ NaOH}(\text{aq})$ were mixed in an insulated polystyrene cup such that the total volume was always 60 cm^3 .

The initial temperature of both solutions was 18.0°C . The maximum temperature reached for each mixture was recorded and the results plotted on the graph shown.

Two straight lines of best fit were drawn and extrapolated to intersect at the maximum.

What is the molar enthalpy change of neutralisation (per mole of water formed) for this reaction, in kJ mol^{-1} ?

(Assume that the specific heat capacity of all solutions is $4.0 \text{ J g}^{-1}^\circ\text{C}^{-1}$, the density of dilute solutions is 1.0 g cm^{-3} , and all heat transferred stays in the solution.)

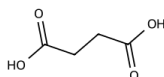


- A -2.4 kJ mol^{-1}
- B -30 kJ mol^{-1}
- C -40 kJ mol^{-1}
- D -60 kJ mol^{-1}
- E -120 kJ mol^{-1}
- F $-2400 \text{ kJ mol}^{-1}$

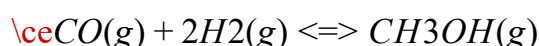
- 13** A sample of 1.18 g of the dicarboxylic acid shown below is reacted with an excess of aqueous sodium hydrogencarbonate ($\text{NaHCO}_3(\text{aq})$) at room temperature and pressure.

Assuming that 1 mol of gas occupies 24.0 dm^3 under these conditions, what is the volume of gas collected?

(A_r values: $\text{H} = 1.0$, $\text{C} = 12.0$, $\text{O} = 16.0$, $\text{Na} = 23.0$)



- A** 0.24 cm^3
B 0.48 cm^3
C 120 cm^3
D 240 cm^3
E 480 cm^3
F 960 cm^3
- 14** Methanol can be manufactured industrially by the reversible reaction of carbon monoxide with hydrogen in the presence of a catalyst:



In a closed reaction vessel at constant temperature and pressure, a mixture containing 8.00 mol of $\text{CO}(\text{g})$ and 20.0 mol of $\text{H}_2(\text{g})$ is allowed to reach equilibrium. At equilibrium, 75.0% of the CO has been converted to $\text{CH}_3\text{OH}(\text{g})$.

What is the total volume of the resulting equilibrium gas mixture?

(Assume that under these conditions, one mole of gas occupies a volume of 45.0 dm^3 .)

- A** 270 dm^3
B 450 dm^3
C 540 dm^3
D 630 dm^3
E 720 dm^3
F 990 dm^3
G 1260 dm^3

- 15** The table shows the melting points and electrical conductivities of four pure substances, P, Q, R, and S, at standard atmospheric pressure:

Substance	Melting point / °C	Electrical conductivity of solid	Electrical conductivity of liquid
P	-78	non-conductor	non-conductor
Q	770	non-conductor	good conductor
R	1420	non-conductor	non-conductor
S	1538	good conductor	good conductor

Which of the following statements is/are correct?

- 1 Melting substance P requires breaking strong covalent bonds between atoms.
- 2 The high melting point of substance R is due to the breaking of strong covalent bonds in a giant lattice.
- 3 In molten Q, electrical conduction is due to mobile ions, whereas in solid S, it is due to delocalised electrons.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 16** Element Q is in Group 17, Period 4 of the Periodic Table. It exists naturally as a mixture of two isotopes with mass numbers 79 and 81. The relative atomic mass of element Q is 79.9.

Which of the following statements is/are correct about element Q?

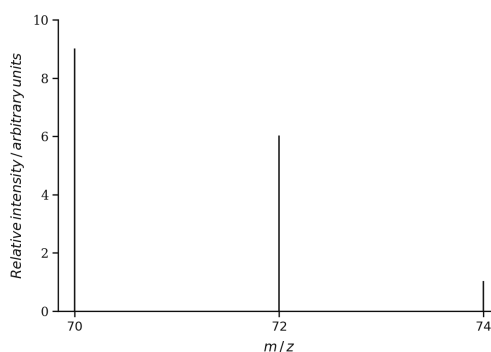
1. 55% of the atoms in a sample of element Q have a mass number of 79.
2. Aqueous element Q oxidises chloride ions to chlorine.
3. Element Q reacts with calcium to form a compound with the empirical formula CaQ_2 that conducts electricity when molten.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 17 The mass spectrum of the molecular ion region for a sample of a pure diatomic gas, X_2 , is shown below.

Element X reacts with phosphorus to form the trihalide PX_3 . Naturally occurring phosphorus consists exclusively of the single stable isotope ^{31}P .

How many peaks appear in the molecular ion region of the mass spectrum of PX_3 , and what is the ratio of the height of the lightest molecular ion peak to that of the heaviest molecular ion peak?



- A 3 peaks; ratio 9 : 1
- B 3 peaks; ratio 27 : 1
- C 4 peaks; ratio 3 : 1
- D 4 peaks; ratio 9 : 1
- E 4 peaks; ratio 27 : 1
- F 4 peaks; ratio 81 : 1
- G 6 peaks; ratio 27 : 1
- H 8 peaks; ratio 27 : 1

- 18** A solid element **M** burns in air with an intense, bright white flame to form a single white solid oxide.

This oxide reacts completely with a dilute aqueous solution of acid **A** to form a colourless solution containing a single salt.

When aqueous barium chloride is added to a sample of acid **A**, followed by dilute hydrochloric acid, a dense white precipitate remains.

When aqueous sodium hydroxide is added dropwise to the solution of the salt formed from the oxide of **M** and acid **A**, a white precipitate forms which does not dissolve in excess sodium hydroxide.

Acid **A** also reacts with an insoluble green solid carbonate **Q** to produce three products, one of which is a blue solution.

Which row in the table correctly identifies **M**, **A**, and **Q**?

	Element M	Acid A	Carbonate Q
A	Ca	HCl	CuCO_3
B	Ca	H_2SO_4	FeCO_3
C	Mg	HCl	CaCO_3
D	Mg	HNO_3	CuCO_3
E	Mg	H_2SO_4	CaCO_3
F	Mg	H_2SO_4	CuCO_3
G	Zn	HNO_3	FeCO_3
H	Zn	H_2SO_4	CuCO_3

- 19** An excess of dilute nitric acid was added to a sample of solid calcium carbonate in an open conical flask placed on an electronic balance:



The total mass of the flask and its contents was recorded at the start ($t = 0$ s) and at 10 s intervals thereafter.

The reaction was complete (stopped) exactly at $t = 40$ s.

Which row in the table could represent the readings displayed on the electronic balance during this experiment?

	0 s	10 s	20 s	30 s	40 s	50 s
A	200.0 g	197.6 g	196.4 g	195.8 g	195.6 g	195.6 g
B	200.0 g	197.6 g	196.4 g	195.8 g	195.6 g	195.5 g
C	200.0 g	197.4 g	196.0 g	195.6 g	195.6 g	195.6 g
D	200.0 g	198.9 g	197.8 g	196.7 g	195.6 g	195.6 g
E	200.0 g	199.8 g	199.2 g	198.0 g	195.6 g	195.6 g
F	200.0 g	202.4 g	203.6 g	204.2 g	204.4 g	204.4 g

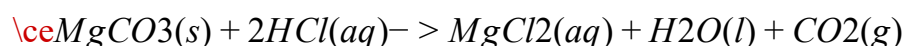
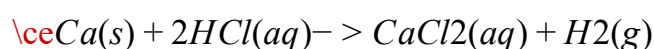
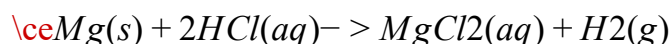
- 20** Three open beakers, each containing an excess of dilute hydrochloric acid, HCl(aq) , are placed on separate digital balances.

The initial mass reading on each balance is recorded.

Equal masses (5.00 g) of three pure solids are added separately to the beakers:

- Beaker 1: magnesium metal, Mg(s)
- Beaker 2: calcium metal, Ca(s)
- Beaker 3: magnesium carbonate, $\text{MgCO}_3\text{(s)}$

In each beaker, a reaction occurs and the gas produced escapes freely into the atmosphere:



Assume that all reactions go to completion, no liquid splashes out, and evaporation of water is negligible.

[A_r values: $\text{H} = 1.0$; $\text{C} = 12.0$; $\text{O} = 16.0$; $\text{Mg} = 24.3$; $\text{Ca} = 40.1$]

Which of the following statements is/are correct?

- 1 The final balance reading for each beaker is greater than its initial reading before the solid was added.
- 2 Beaker 1 produces the greatest volume of gas, measured at room temperature and pressure.
- 3 Beaker 1 loses the greatest mass of gas to the surroundings.

- A** none
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

21 Which of the following species have all of their atoms lying in the same plane?



A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

22 The hydrolysis of one mole of triglyceride into glycerol and fatty acids has an enthalpy change of $\Delta H = +15 \text{ kJ mol}^{-1}$. The table gives the bond energies for the bonds involved, except for the ester C-O bond.

Bond	Bond Energy / kJ mol^{-1}
O-H (water)	464
O-H (glycerol)	455
C-O (fatty acid)	358

Assume that the hydrolysis of each ester linkage involves breaking one ester C-O bond and one water O-H bond, and forming one fatty acid C-O bond and one glycerol O-H bond.

What is the bond energy of the ester C-O bond?

A 344 kJ mol^{-1}

B 354 kJ mol^{-1}

C 362 kJ mol^{-1}

D 363 kJ mol^{-1}

E 364 kJ mol^{-1}

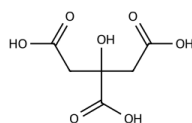
- 23** A 1.60 g sample containing compound **X** and an unreactive impurity was added to an excess of dilute aqueous sodium hydrogencarbonate, $\text{NaHCO}_3(\text{aq})$.

The structure of compound **X** is shown below.

The volume of carbon dioxide gas collected at room temperature and pressure was 360 cm^3 .

What is the mass of the impurity in the sample?

(A_r : $\text{H} = 1.0$, $\text{C} = 12.0$, $\text{O} = 16.0$; molar gas volume at room temperature and pressure = $24\,000 \text{ cm}^3 \text{ mol}^{-1}$)



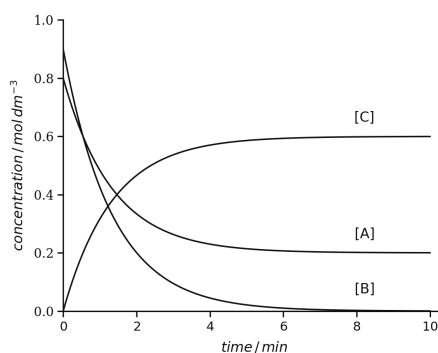
- A** 0.16 g
- B** 0.64 g
- C** 0.72 g
- D** 0.88 g
- E** 0.96 g
- F** 1.44 g

- 24 Substances A and B react in a closed container of fixed volume to form substance C . The reaction goes to completion.

The graph shows how the concentrations of A , B , and C change over time.

Which row in the table correctly gives the balanced equation for the reaction, the limiting reactant, and the percentage of the initial amount of the excess reactant that has reacted?

	Balanced equation	Limiting reactant	Percentage of excess reactant reacted
A	$2A + 3B \rightarrow 2C$	B	75%
B	$2A + 3B \rightarrow 2C$	B	25%
C	$2A + 3B \rightarrow 2C$	A	75%
D	$3A + 2B \rightarrow 2C$	B	75%
E	$3A + 2B \rightarrow 2C$	A	25%
F	$4A + 3B \rightarrow 2C$	B	75%
G	$4A + 3B \rightarrow 2C$	A	25%
H	$A + 3B \rightarrow 2C$	B	33%



- 25** Gas-liquid chromatography with a non-polar stationary phase was used to analyse three hydrocarbon mixtures labelled Sample X, Sample Y, and Sample Z.

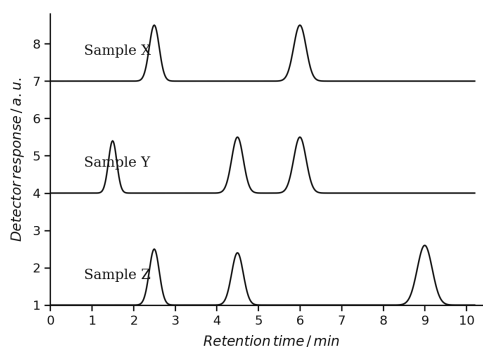
The table gives the retention times of five pure reference hydrocarbons obtained under the same experimental conditions:

Hydrocarbon	Retention time / min
methylpropane	1.5
butane	2.5
2-methylbutane	4.5
pentane	6.0
hexane	9.0

The chromatogram obtained for the three samples is shown.

Which of the following statements is/are correct?

- 1 Sample X contains only unbranched alkanes.
- 2 Butane is present in all three samples.
- 3 Both Sample Y and Sample Z contain at least one hydrocarbon with the molecular formula C_5H_{12} .
- 4 Hexane is the most volatile of the five reference hydrocarbons.



- A** 1 only
- B** 3 only
- C** 1 and 3 only
- D** 1 and 4 only
- E** 2 and 3 only
- F** 2 and 4 only
- G** 1, 3 and 4 only
- H** 1, 2, 3 and 4

- 26** An aqueous solution of barium hydroxide, $\text{Ba}(\text{OH})_2(\text{aq})$, is mixed in an equimolar ratio with dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$. A white precipitate of barium sulfate, $\text{BaSO}_4(\text{s})$, forms alongside liquid water, $\text{H}_2\text{O}(\text{l})$, causing the electrical conductivity of the mixture to drop to near zero.

Which of the following represents the correct balanced ionic equation for this reaction?

- A** $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
- B** $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
- C** $\text{Ba}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- D** $\text{Ba}(\text{OH})_2(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- E** $\text{Ba}^{2+}(\text{aq}) + \text{OH}^-(\text{aq}) + \text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + \text{H}_2\text{O}(\text{l})$
- F** $\text{Ba}(\text{OH})_2(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$

- 27** The monatomic ions X^{2+} , Y^- and Z^{3-} are isoelectronic (they contain the same number of electrons).

Which of the following gives the elements X, Y and Z in order of increasing atomic number (lowest first)?

- A** X, Y, Z
- B** Z, Y, X
- C** Y, X, Z
- D** Z, X, Y